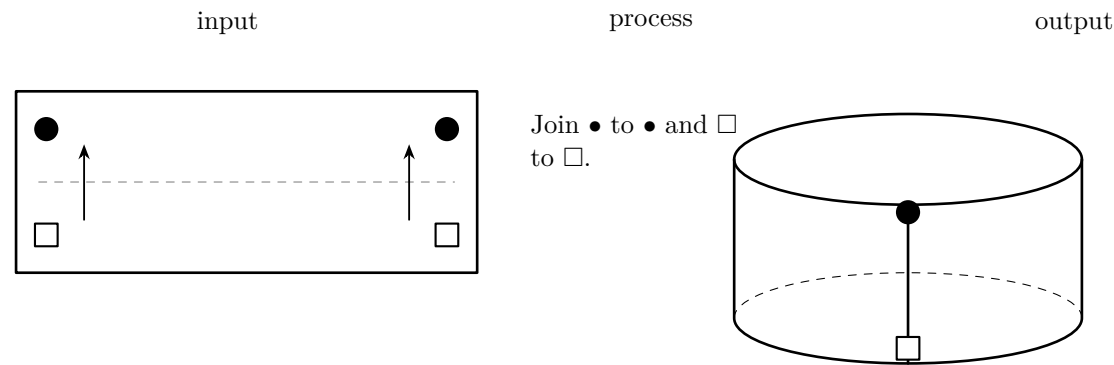
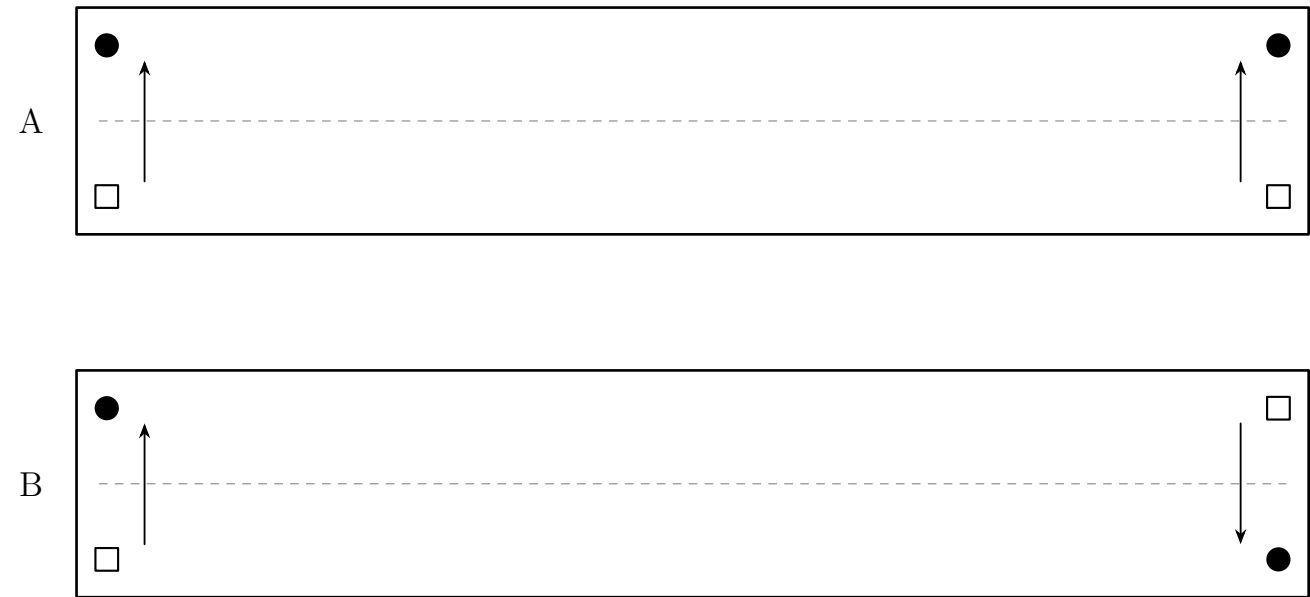


Keep the bands on the table. An adult makes every cut. The rectangles are joining recipes.



An edge trip stays on an edge, follows its seam marks, and stops on its first return to the start dot.

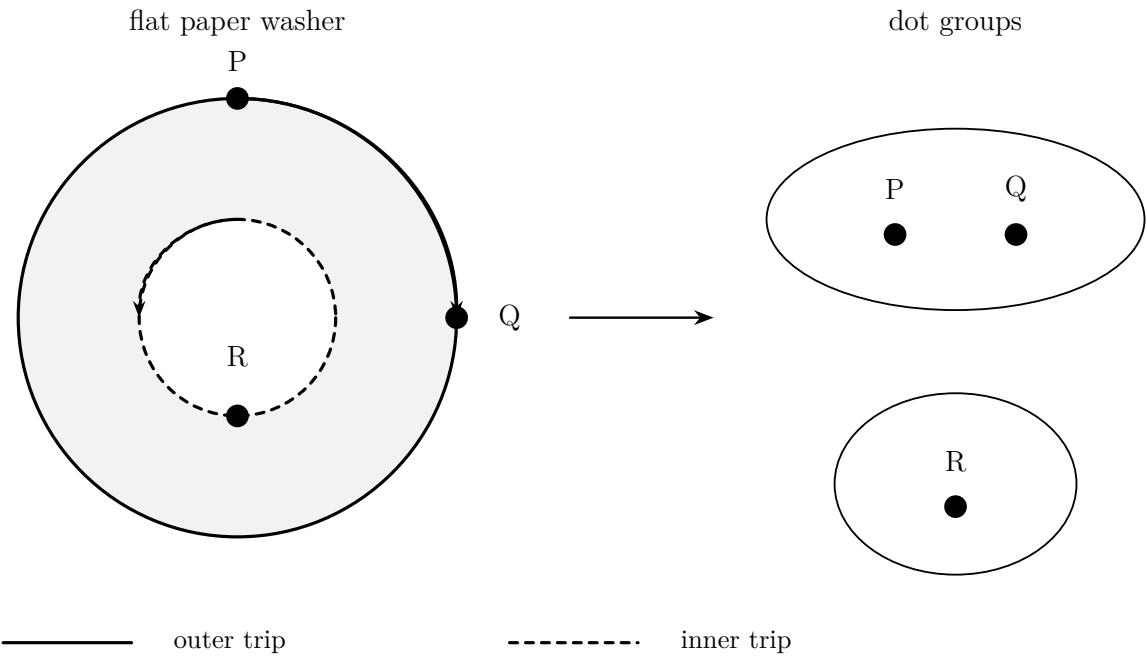
Problem 1: Trace all the edges of A and B. Show on each flat recipe where each separate closed edge goes through the seam.



A: _____ B: _____

Problem 2: One separate closed edge is a boundary component. Explain the number of boundary components of A and B from their seam recipes.

Dots on one edge go in one ring.

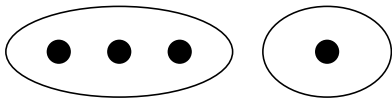


The rings record which dots share an edge.

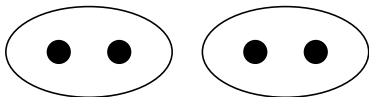
Problem 3: Place four edge dots to match each picture on A and on B, or cross it out. Each ring means a different closed edge; other edges may have no dots.



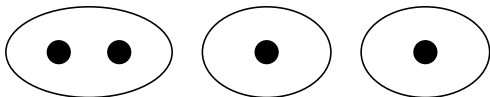
A: _____ B: _____



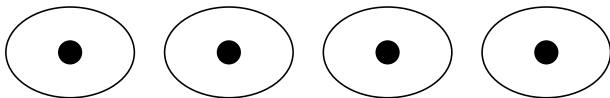
A: _____ B: _____



A: _____ B: _____

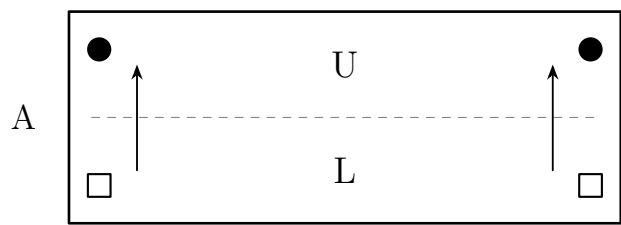


A: _____ B: _____



A: _____ B: _____

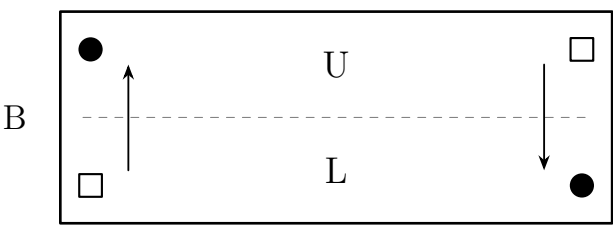
Problem 4: Predict the number of connected pieces after a middle-line cut in A and B. Have an adult cut both bands, then count the pieces and the boundary components of each piece.



I predict: _____



after cutting A

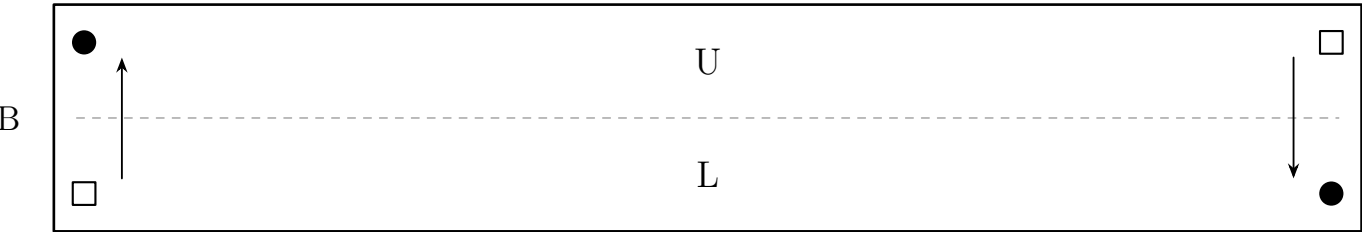
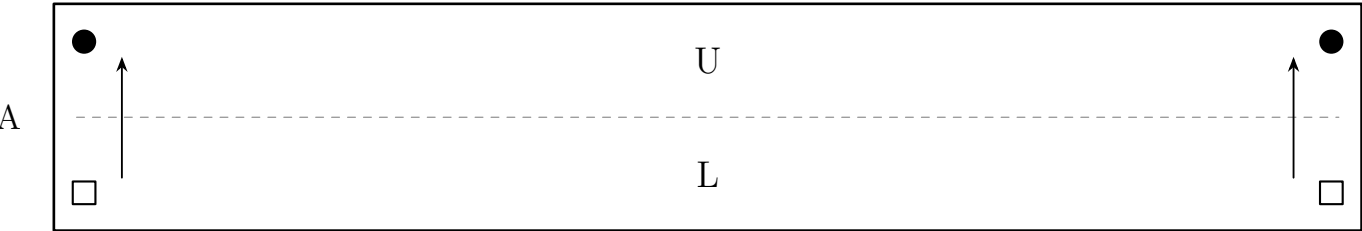


I predict: _____

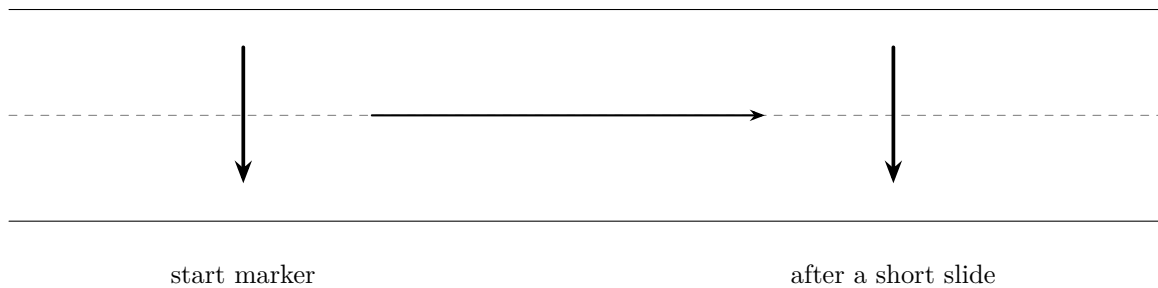


after cutting B

Problem 5: Use U and L to explain the connected pieces after the cuts. Explain separately the number of boundary components on each piece.



Problem 6: Ask an adult to rebuild uncut A and B from the two spare strips. Put a removable arrow across each strip, pointing from U toward L. Slide it along the middle line for one full trip, keeping it across the strip.



Compare the returning arrow with its starting position. Explain its return direction from the seam.

Problem 7: Which four-dot pictures from Problem 3 are possible after each cut? Explain how your answer separates connected pieces from boundary components.